

```
// Experiment 3: Interfacing Potentiometer for Position Control  
Simulation and Display on LCD
```

```
#include <Wire.h>  
#include <LiquidCrystal_I2C.h>  
#include <Servo.h>
```

```
LiquidCrystal_I2C lcd(0x27, 16, 2);  
const int Pot = A0;  
Servo myservo;
```

```
void setup() {  
    pinMode(Pot, INPUT);  
    myservo.attach(8);  
  
    Serial.begin(9600);  
    Serial.println("Potentiometer to Servo Angle Control");  
    Serial.println("=====");  
  
    lcd.init();  
    lcd.backlight();  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Pot: ");  
    lcd.setCursor(0, 1);  
    lcd.print("Angle: ");
```

```
}

void loop() {
    int potVal = analogRead(Pot);
    int angle = map(potVal, 0, 1023, 0, 180);
    myservo.write(angle);

    Serial.print("Potentiometer Value: ");
    Serial.print(potVal);
    Serial.print("    |    Servo Angle: ");
    Serial.println(angle);

    lcd.setCursor(5, 0);
    lcd.print(potVal);
    lcd.print("    ");
    lcd.setCursor(7, 1);
    lcd.print(angle);
    lcd.print((char)223);
    lcd.print("    ");

    delay(100);
}
```